

|                |                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | $-1.12 \times 10^{-11} \text{ N}$                                                                                                                                                        |
| <b>7(a)(i)</b> | Random – Through experiments, the radioactive decay is unpredictable and therefore the count rate in experiment is from average of all radioactive samples.                              |
| <b>7(a)(i)</b> | Spontaneous – No external stimuli is required to cause radioactive decay.                                                                                                                |
| <b>7(b)(i)</b> | $A = A_0 e^{-\lambda t}$ <p>Choose random points from graph</p> $4 \times 10^5 = 5.8 \times 10^5 e^{-\lambda(8 \times 24 \times 3600)}$ $\lambda_1 = 5.38 \times 10^{-7} \text{ s}^{-1}$ |

|          |                                                                                                                                                                                                             |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | $2 \times 10^5 = 5.8 \times 10^5 e^{-\lambda(23 \times 24 \times 3600)}$ $\lambda_2 = 5.358 \times 10^{-7} \text{ s}^{-1}$ $\lambda = \frac{\lambda_1 + \lambda_2}{2} = 5.37 \times 10^{-7} \text{ s}^{-1}$ |
| 7(b)(ii) | $A = \lambda N$ $5.8 \times 10^5 = (5.3 \times 10^{-7}) N$ $N = 1.0943 \times 10^{12} = 1.818 \times 10^{-12} \text{ mol}$ $\text{mass} = 5.82 \times 10^{-11} \text{ g}$                                   |
| 8(a)     | Internal energy is the sum of the microscopic Kinetic Energy and Microscopic                                                                                                                                |